

**TABLE I**  
**POPULAR OPERATING SYSTEMS**  
**AMONG THE IoT COMMUNITY**

| Operating system | Features                                                                                                                                                                                                                                               |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Contiki          | <ul style="list-style-type: none"> <li>• 10kB RAM and 30kB flash required</li> <li>• Extensive wireless networking support with an IPv6 stack</li> <li>• Touted for efficient memory allocation</li> </ul>                                             |
| RIOT OS          | <ul style="list-style-type: none"> <li>• 1.5kB RAM and 5kB flash required</li> <li>• Multi-threading, dynamic memory management, hardware abstraction, partial POSIX compliance and C++ support</li> <li>• Touted for efficient power usage</li> </ul> |
| TinyOS           | <ul style="list-style-type: none"> <li>• Smallest operating system with the base less than 400 bytes</li> <li>• Most applications fit in 16kB</li> </ul>                                                                                               |
| Raspbian         | <ul style="list-style-type: none"> <li>• Popular with Raspberry Pi DIYers</li> <li>• Supports Node-RED</li> <li>• Linux based</li> </ul>                                                                                                               |
| Brillo           | <ul style="list-style-type: none"> <li>• 32MB RAM and 128MB flash required</li> <li>• High adoption among hacker boards</li> <li>• Tied to Google's Weave communications protocol built on Android</li> </ul>                                          |

kilometres," he explains. Post approval from WPC, this would be an interesting localised GPS to look out for.

**Narrowband for the machines.** Narrowband Long Term Evolution (NB-LTE), for example, can be fully integrated into existing LTE networks

without the need of an overlay network. A version of LTE is being tested out for M2M communication as well. LTE machine-type communication (LTE-MTC), or LTE-M, also works within the normal constructs of LTE networks. However, it is

more energy efficient in terms of its extended discontinuous repetition cycle, meaning that the endpoint can communicate with the access point on how often it listens for the downlink.

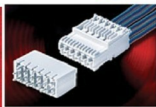
The advantages do not end here though. Applications are extremely simple as well. A mobile phone operator only has to upload new base-band software onto its base station, hence saving up heavily on added antenna costs.

It is also expected to be simpler because of the operation in the 1.4MHz band of the channel instead of the 20MHz in regular communication. This, however, has been further simplified into NB-LTE-M that dedicates existing resource blocks to the IoT traffic, by using further smaller channels. Instead of using 1.4MHz spectrum and six resource blocks, it works with only one LTE resource block, giving an effective throughput of about 200kbps for downlink and 144kbps for uplink.



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